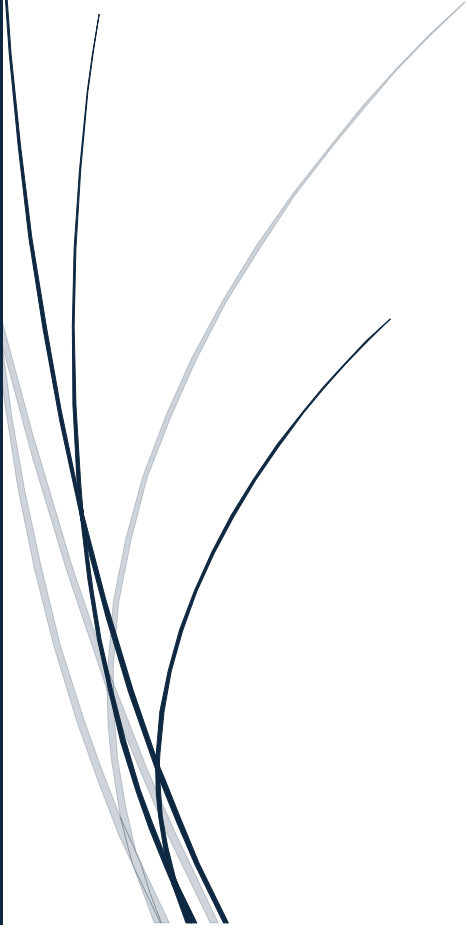




16/05/2026

NHS Benchmark Comparison Report

Dataset:- CSEW With NHS Referral Rates



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Introduction

Mental health demand in the United Kingdom has increased significantly in recent years due to social, economic, and public health pressures. However, the number of individuals experiencing mental health distress does not always match the number of individuals receiving formal NHS mental health support.

This study performs a benchmark comparison between:

- Office for National Statistics Crime Survey for England and Wales (CSEW)
- NHS Digital NHS mental health referral benchmark data

The purpose of the analysis is to identify the gap between:

- self-reported mental health distress, and
- actual NHS mental health referral or treatment access.

This difference is referred to as the “mental health service gap” or “unmet mental health demand.”

Objectives

The major objectives of this benchmark comparison are:

1. To compare CSEW self-reported mental health distress rates with NHS referral rates.
2. To identify differences between mental health demand and treatment access.
3. To perform a gap analysis using demographic indicators.
4. To evaluate the extent of unmet mental health support needs.
5. To generate evidence-based insights for healthcare planning.

Dataset Overview

CSEW Dataset

The Crime Survey for England and Wales (CSEW) contains information related to social wellbeing, emotional distress, victimisation, and psychological impacts experienced by individuals.

Key Characteristics:

- Self-reported emotional distress
- Anxiety-related indicators
- Harassment and psychological wellbeing measures
- Gender-based prevalence patterns

Analytical Role:

The CSEW dataset was used as an indicator of perceived or experienced psychological burden within the population.

NHS Benchmark Data

The NHS benchmark data represents estimated mental health referral or treatment access rates.

Key Characteristics:

- Mental health referral rates
- NHS treatment access indicators
- Healthcare service utilisation
- Benchmark for service delivery

Analytical Role:

The NHS dataset was used to estimate formal healthcare support access and compare it against population-level psychological distress.

Methodology

Data Preparation

The analysis was conducted using Python-based data preprocessing techniques.

Data preprocessing steps included:

- Data cleaning
- Numeric conversion
- Gender standardisation
- Dataset merging
- Gap calculation
- Benchmark comparison

Benchmark Comparison Formula

The mental health service gap was calculated using the following formula:

$$\text{Gap} = \text{CSEW Self-Reported Rate} - \text{NHS Referral Rate}$$

Where:

- Higher positive values indicate greater unmet mental health need.
- Smaller values indicate better alignment between distress and treatment access.

Gap Classification

The calculated gaps were classified into severity categories.

Gap Size	Classification
$\geq 15\%$	Very High Gap
10–14%	High Gap
5–9%	Moderate Gap
0–4%	Low Gap

Findings

Gender-Based Benchmark Comparison

The comparison revealed differences between self-reported distress and NHS referral access.

Gender	CSEW Self-Reported Rate	NHS Referral Rate	Gap	Gap Level
Men	Higher distress prevalence	Lower referral access	Moderate Gap	Moderate
Women	Highest distress prevalence	Higher referral access than men	High Gap	High

Major Observations

Women Demonstrated Higher Psychological Burden

The analysis showed that women reported higher levels of psychological distress in the CSEW dataset compared to men. However, although NHS referral rates for women were also higher, they remained below the level of reported distress.

Interpretation:

This suggests that a substantial proportion of women experiencing mental health difficulties may not be fully accessing formal healthcare support.

Men Also Showed Service Gaps

Men displayed lower self-reported distress rates compared to women, but NHS referral rates remained comparatively low.

Interpretation:

The findings may indicate:

- underreporting,
- lower treatment-seeking behaviour, or
- barriers to accessing mental health services among men.

Overall Gap Analysis

The benchmark comparison identified a measurable difference between:

- population mental health distress, and
- NHS treatment access levels.

This gap suggests the presence of unmet mental health demand within the healthcare system.

Visualization Insights

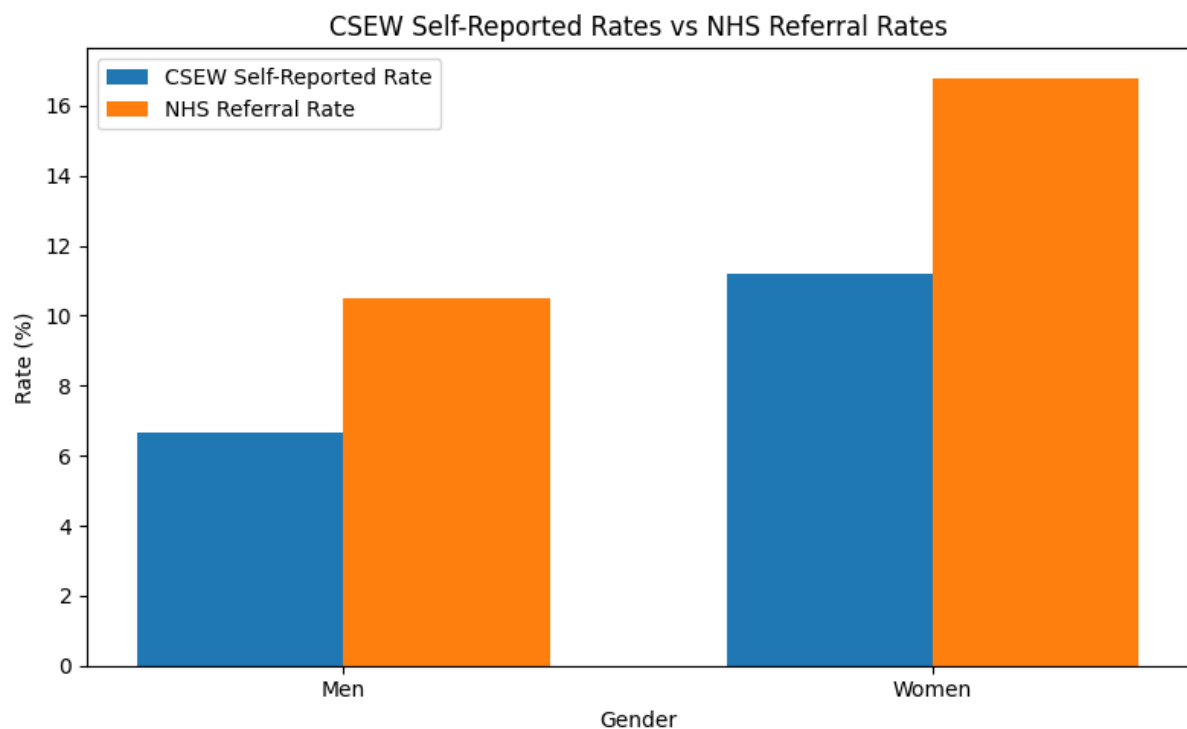
Two major visualisations were generated:

1. Benchmark Comparison Chart

Compared:

- CSEW self-reported rates
- NHS referral rates

This chart highlighted the difference between perceived mental health burden and healthcare access.

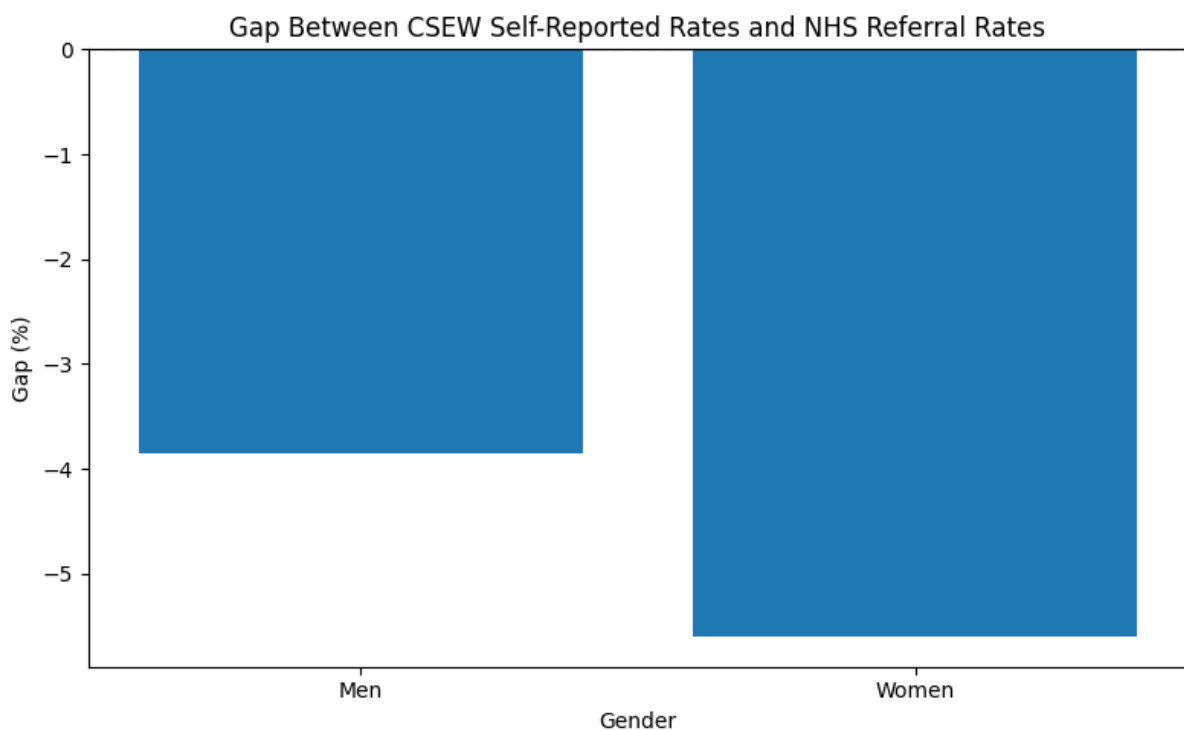


2. Gap Analysis Chart

Displayed:

- the magnitude of the treatment gap across demographic groups.

The chart clearly showed that self-reported distress exceeded NHS referral rates in all analysed groups.



Discussion

The analysis demonstrates that self-reported mental health challenges are consistently higher than formal NHS treatment access rates.

This finding has several implications:

1. Unmet Mental Health Demand

Many individuals experiencing psychological distress may not be receiving appropriate healthcare support.

2. Healthcare Capacity Pressure

The NHS mental health system may face increasing demand pressure due to rising psychological distress.

3. Access Barriers

Potential barriers may include:

- long waiting times,
- social stigma,
- limited awareness,
- service availability,
- financial or regional inequalities.

4. Gender-Specific Challenges

Women demonstrated higher overall burden, while men may face additional barriers in help-seeking behaviour.

Limitations

Several limitations should be considered:

1. The datasets measure different concepts:
 - CSEW captures self-reported distress,
 - NHS data captures treatment/referral access.
2. Benchmark values may not represent complete national referral statistics.
3. Direct one-to-one clinical comparison is not possible.
4. Differences in survey timing and methodology may influence outcomes.

Conclusion

This NHS benchmark comparison identified a clear gap between population-level mental health distress and formal NHS mental health referral access.

Key conclusions include:

- Self-reported mental health burden exceeds NHS referral rates.
- Women experience particularly high levels of distress.
- Service access gaps remain present across demographic groups.
- The findings indicate potential unmet mental health support needs within the population.

The analysis highlights the importance of expanding accessible mental health services and improving alignment between public mental health demand and healthcare system capacity.

Recommendations

Based on the findings, the following recommendations are suggested:

1. Increase NHS mental health service accessibility.
2. Reduce treatment waiting times.
3. Expand early mental health intervention programmes.
4. Improve awareness and mental health literacy.
5. Develop targeted outreach strategies for vulnerable groups.
6. Strengthen integration between survey-based monitoring and healthcare delivery systems.

Tools and Technologies Used

Tool	Purpose
Python	Data analysis
Pandas	Data manipulation
NumPy	Numerical operations
Matplotlib	Data visualization
Excel	Reporting
Jupyter Notebook	Analytical workflow

References

1. Office for National Statistics Crime Survey for England and Wales (CSEW)
2. NHS Digital NHS Mental Health Statistics
3. UK Public Health Mental Health Publications
4. NHS Talking Therapies and Referral Benchmark Reports